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Find a non-isomorphic tree with five vertices.

VuAnswers.com

Answer ([Please click here to Add Answer](#))

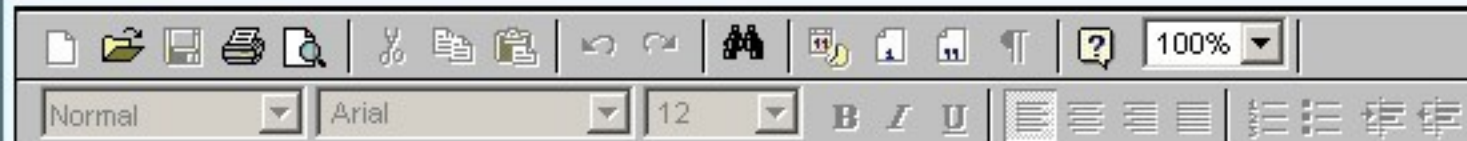


Made by: Waqar Siddhu

Construct input/output table for $P \wedge q$

VuAnswers.com

Answer ([Please click here to Add Answer](#))



Made by: Waqar Siddhu

If there are 3 different optional indoor games in 6 different optional outdoor games for students in sports week then find the number of choices for a student who wants to select one optional game.

Answer ([Please click here to Add Answer](#))

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Made by: Waqar Siddhu

Find the expectation μ of the distribution given in the following table.

x_i	1	3	4	5
$f(x_i)$	0.1	0.2	0.3	0.4

Answer ([Please click here to Add Answer](#))

VuAnswers.com

Rich text editor toolbar with icons for file operations, undo/redo, bold, italic, underline, list, and link, along with a 100% zoom level.

Made by: Waqar Siddhu

- i) Evaluate $P(5, 3)$
- ii) How many 5-permutations are there of a set of five objects?

VuAnswers.com

Answer ([Please click here to Add Answer](#))**Made by: Waqar Siddhu**

Assume that for the truth values $p = F$, $q = T$, $r = F$. Show that the proposition : $p \vee (q \wedge r)$ is true.

VuAnswers.com

Answer ([Please click here to Add Answer](#))



Made by: Waqar Siddhu

Find the variance σ^2 of the distribution given in the following table.

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Answer ([Please click here to Add Answer](#))

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Rich text editor toolbar with icons for file operations, undo/redo, bold, italic, underline, list, and link, along with a font face dropdown (Normal, Arial) and a font size dropdown (12).

Made by: Waqar Siddhu

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Answer ([Please click here to Add Answer](#))

VuAnswers.com

**Made by: Waqar Siddhu**

Determine whether the following graph has Hamiltonian circuit, justify your answer.

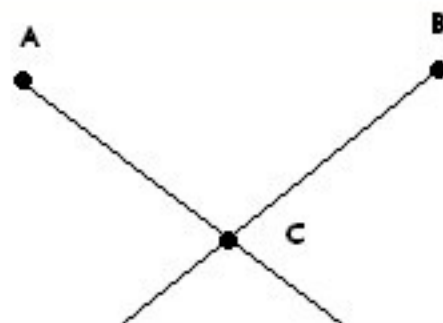


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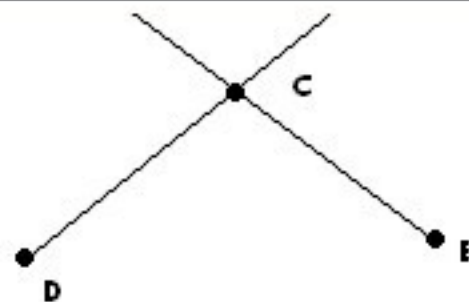


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Answer ([Please click here to Add Answer](#))

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Made by: Waqar Siddhu

Prove that $A - B = A \cap B^c$ by using Membership Table.

VuAnswers.com

Answer ([Please click here to Add Answer](#))



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Find the GCD of 11425,450 using Division Algorithm.

VuAnswers.com

Answer ([Please click here to Add Answer](#))



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Using handshaking theorem show that the maximum number of edges in a complete graph with n vertices is $\frac{n(n-1)}{2}$.

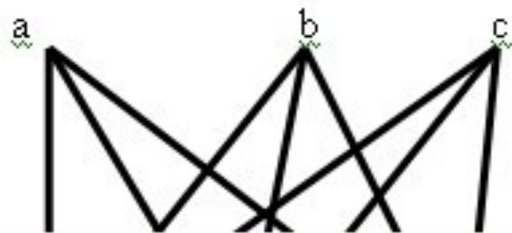
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Determine whether the graph is planer. If so, draw it such that no edges cross.

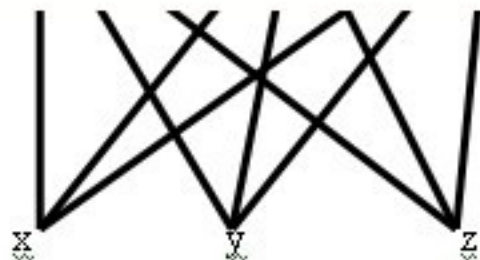


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Answer ([Please click here to Add Answer](#))

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Made by: Waqar Siddhu

Let $A = \{1, 2, 3\}$ be a set, R and S be the following relations on it

$$R = \{(1, 2), (1, 3), (2, 3), (3, 1), (3, 3)\}$$

$$S = \{(1, 2), (1, 3), (2, 1), (3, 3)\}$$

Find $R \cap S$.

Answer ([Please click here to Add Answer](#))

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A cafeteria offers a choice of two soups, five sandwiches, three desserts and three drinks. How many different lunches, each consisting of a soup, a sandwich, a dessert and a drink are possible?

Answer ([Please click here to Add Answer](#))

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Made by: Waqar Siddhu

Find the expectation μ of the distribution given in the following table.

x_i	1	3	4	5
$f(x_i)$	0.1	0.2	0.3	0.4

Answer ([Please click here to Add Answer](#))

VuAnswers.com



Made by: Waqar Siddhu

Let $A = \begin{bmatrix} 1 & 3 & 7 \\ 5 & 2 & 9 \end{bmatrix}$ then find A^t .

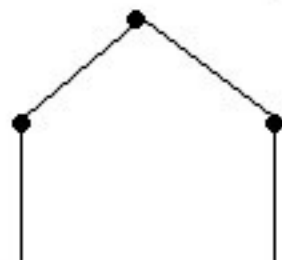
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Made by: Waqar Siddhu

Determine the chromatic number of the given graph by inspection.

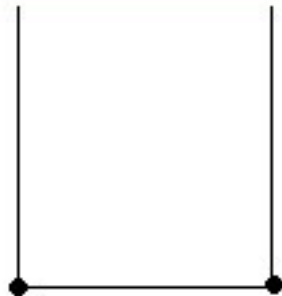


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let A and B are two sets then show that $B \subseteq A \cup B$.

VuAnswers.com

Answer ([Please click here to Add Answer](#))



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The members of a club are 12 boys and 8 girls. In how many ways can a committee of 3 boys and 2 girls be formed?

Answer ([Please click here to Add Answer](#))

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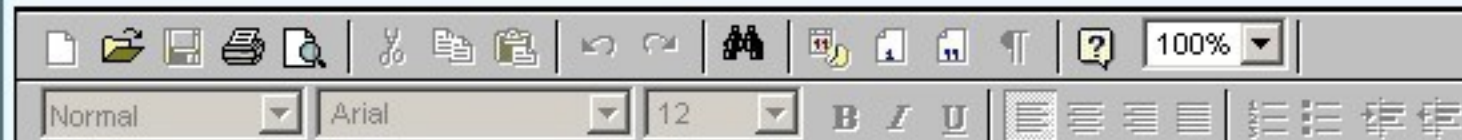


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Determine the probability of appearing an odd number when a fair die is thrown.

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Answer ([Please click here to Add Answer](#))



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Define a binary relation R from $A = \{1, 3\}$ to $B = \{1, 2, 3\}$ as follows

$$R = \{(a, b) \in A \times B \mid a < b\}$$

- (a) Find the ordered pairs in R .
- (b) Find the domain and range of R .
- (c) Is $1R2$ and $3R3$?

Answer ([Please click here to Add Answer](#))

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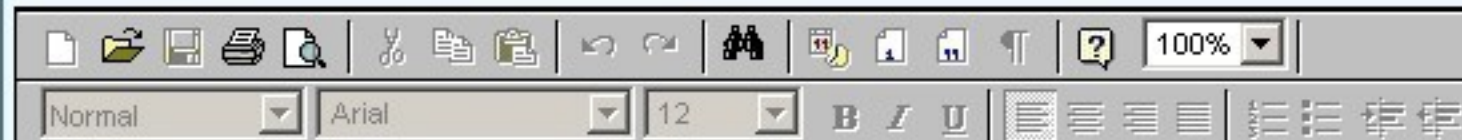


Made by: Waqar Siddhu

A college has 10 basketball players. A 5-member team and a captain will be selected out of these 10 players. How many different selections can be made?

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Answer ([Please click here to Add Answer](#))



Made by: Waqar Siddhu

Find the probability distribution of the sum of the dots when two fair dice are thrown.

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Answer ([Please click here to Add Answer](#))



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Find the product AB and BA of the matrices (if not possible then give reason).

$$A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 2 & 0 & -4 \\ 3 & -2 & 6 \end{bmatrix}$$

Answer ([Please click here to Add Answer](#))

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Find the double negation of the proposition
 $p = \text{It is cold.}$

Answer ([Please click here to Add Answer](#))

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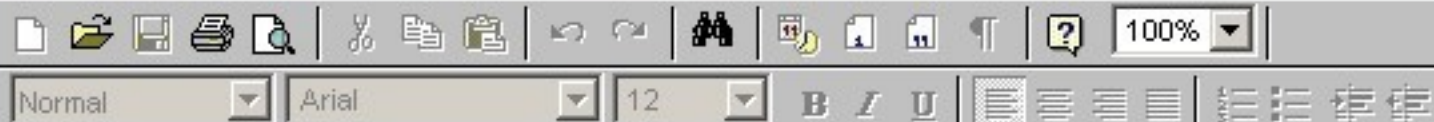


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How many signals can be given by 5 flags of different colors using 3 flags at a time?

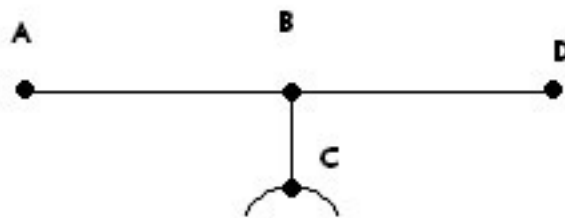
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Find the degree of each vertex in the figure (given below).



Answer ([Please click here to Add Answer](#))

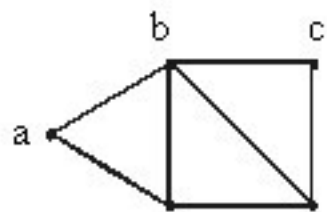
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Find the degree sequence of the following graph.



Answer ([Please click here to Add Answer](#))

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Draw a graph with six vertices, five edges that is not a tree.

Answer ([Please click here to Add Answer](#))

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Let a and b be integers. Suppose a function Q is defined recursively as follows:

$$Q(a,b) = \begin{cases} 5 & \text{if } a < b \\ Q(a-b, b+2) + a & \text{if } b \leq a \end{cases}$$

Find Q (14, 3).

Answer ([Please click here to Add Answer](#))

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Out of five people in an office, just 3 are to be selected to go to an exhibition. In how many ways can the three be chosen?

Answer ([Please click here to Add Answer](#))

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Find the variance σ^2 of the distribution given in the following table.

x_i	1	3	4	5
$f(x_i)$	0.4	0.1	0.2	0.3

Answer ([Please click here to Add Answer](#))

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Let $A = \{3, 6, 9, 12\}$ be a set, find $A \times A$. Determine whether $R = \{(3, 3), (6, 6), (9, 9), (12, 12), (3, 6), (6, 3)\}$ is an equivalence relation on A or not?

Answer ([Please click here to Add Answer](#))

VuAnswers.com



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There are 12 students from whom 5 are going to be chosen to represent their school at a conference. If Jack, Anna or Chris, but only one of them, must be chosen, in how many ways can the students be chosen to go to the conference?

Answer ([Please click here to Add Answer](#))

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Determine whether the given graph has a Hamilton circuit? If it does, find such a circuit, if it does not, give an argument to show why no such circuit exists.

(c)



(b)



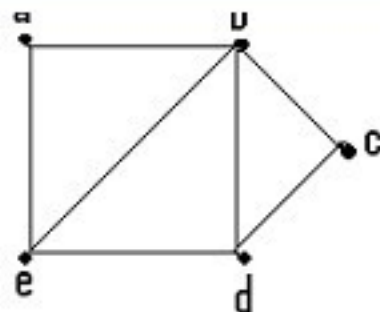
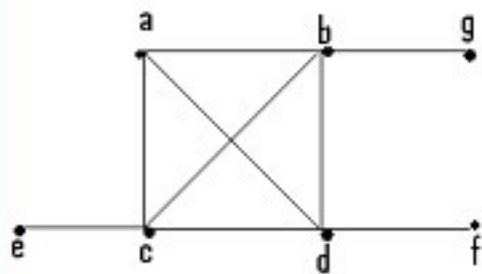
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Answer ([Please click here to Add Answer](#))

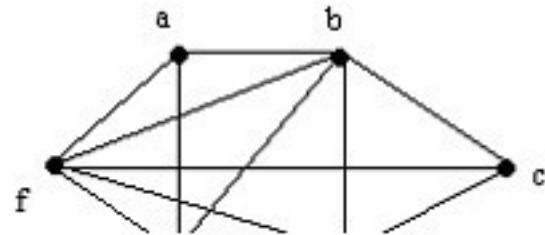
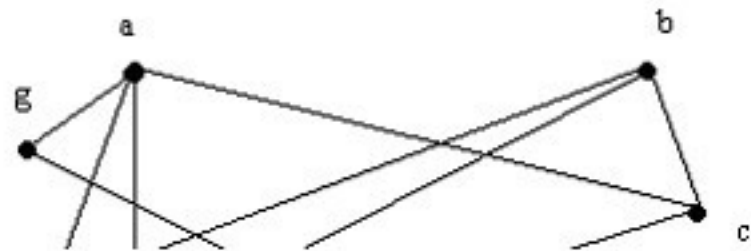
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Consider the following graphs



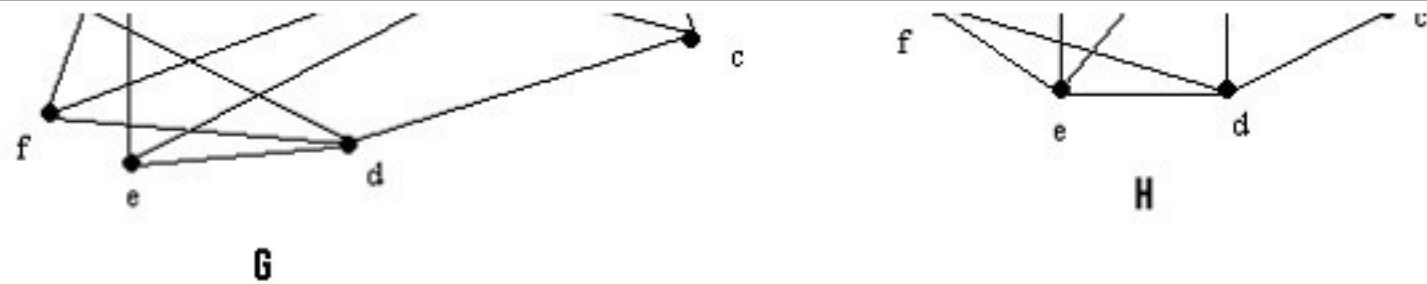
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G

Are the above graphs bipartite? Justify your answer.

Answer ([Please click here to Add Answer](#))

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How many ordered selections of two elements can be made from the set of four elements?

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Answer ([Please click here to Add Answer](#))



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Suppose that f is defined recursively by $f(0) = 1, f(n+1) = 3f(n) + 1$ then find $f(2)$.

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Answer ([Please click here to Add Answer](#))



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Find $P(A|B)$ where

$$P(A) = \frac{1}{2}, P(B) = \frac{1}{3} \text{ and } P(A \cap B) = \frac{1}{4}.$$

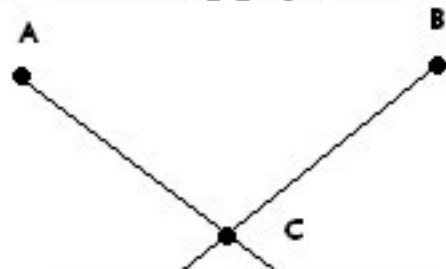
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Answer ([Please click here to Add Answer](#))



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Check whether the following graph has an Euler circuit, justify your answer.

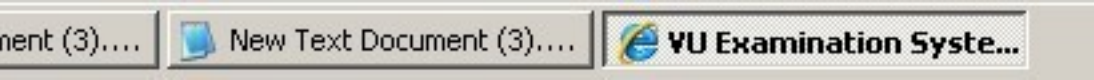
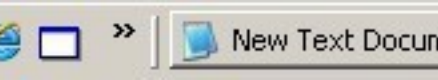


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Answer ([Please click here to Add Answer](#))



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Suppose that p and q are statements so that $q \rightarrow p$ is false. Find the truth values of each of the following:

1. $\sim q \rightarrow p$
2. $p \vee q$
3. $p \leftrightarrow q$

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Answer ([Please click here to Add Answer](#))



Made by: Waqar Siddhu

How many distinguishable ways can the letter of the word HULLABALOO be arranged.

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Answer ([Please click here to Add Answer](#))



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Find the variance σ^2 of the distribution given in the following table.

x_i	1	3	4	5
$f(x_i)$	0.4	0.1	0.2	0.3

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Answer ([Please click here to Add Answer](#))

Rich text editor toolbar with icons for file operations, text formatting, and a 100% zoom level.

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Answer (Please [click here](#) to Add Answer)

Rich text editor toolbar with icons for file operations, text formatting, and a 100% zoom level.

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Determine whether the following graph has Hamiltonian circuit, justify your answer.

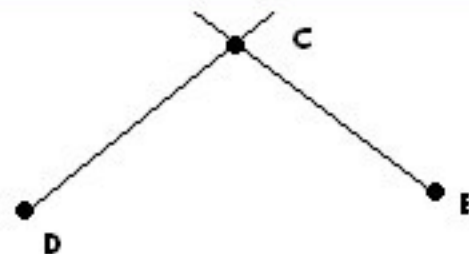


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Answer (Please [click here](#) to Add Answer)

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Use Kruskal's Algorithm to draw the minimal spanning tree for the graph below. Indicate the order in which edges are added to form a tree.

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Answer ([Please click here to Add Answer](#))



Made by: Waqar Siddhu

The fifth term of an arithmetic sequence is 17 and ninth term is 37 find the first four terms of this sequence.

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Answer ([Please click here to Add Answer](#))



Made by: Waqar Siddhu

Find the M number of ways that ten chocolates can be divided among three children if the youngest child is to receive four chocolates and each of the others three chocolates.

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Answer ([Please click here to Add Answer](#))



Made by: Waqar Siddhu

Find all edges which are incident on v_1 , all vertices that are adjacent to v_1 , all loops, all parallel edges and all isolated vertices.

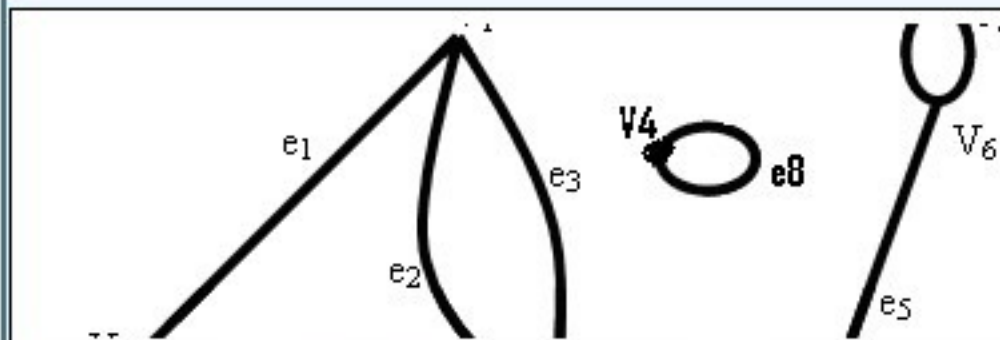


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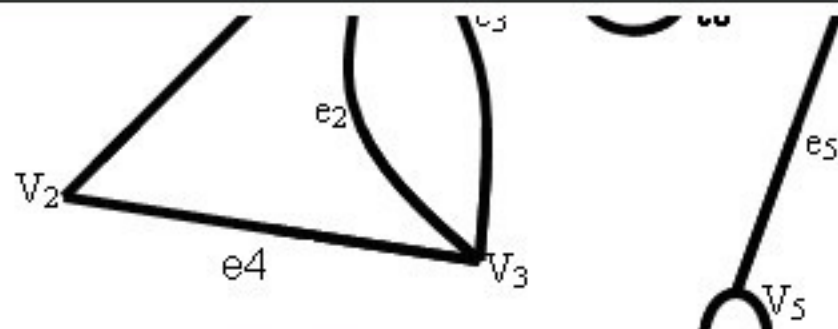
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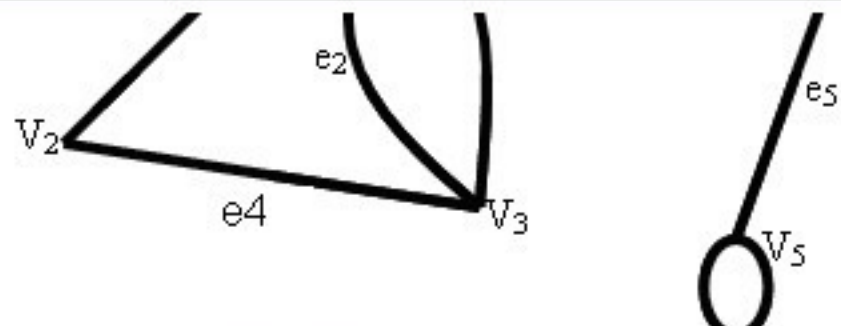
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Answer (Please [click here](#) to Add Answer)

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Suppose that a connected planar simple graph has 20 edges. If a plane drawing of this graph has 10 faces, how many vertices does this graph have?

Answer ([Please click here to Add Answer](#))

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Suppose $A = \{1, 2, 3, 4\}$ and $B = \{x, y, z\}$ are two sets and R is a relation from A to B as $R = \{(1, y), (1, z), (3, y), (4, x), (4, z)\}$ then determine the matrix representation of the R .

Answer ([Please click here to Add Answer](#))

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A cafeteria offers a choice of two soups, five sandwiches, three desserts and three drinks. How many different lunches, each consisting of a soup, a sandwich, a dessert and a drink are possible?

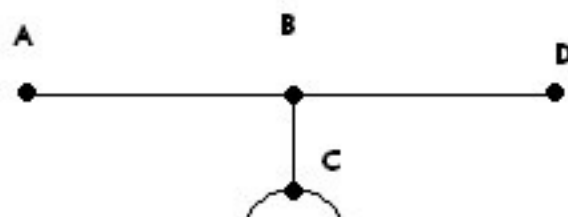
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Find the degree of each vertex in the figure (given below).

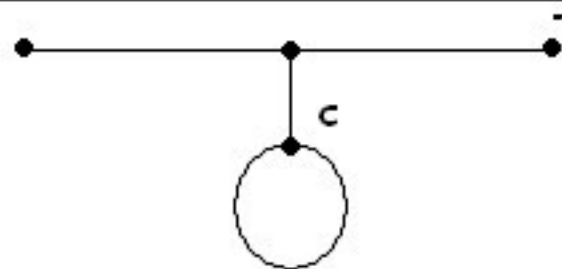


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Answer ([Please click here to Add Answer](#))

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Suppose that R and S are two reflexive relations on a set A . Prove or disprove $R \cap S$ is reflexive.

Answer ([Please click here to Add Answer](#))

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Made by: Waqar Siddhu

- i) Evaluate $P(5, 2)$
- ii) How many 4-permutations are there of a set of seven objects?

Answer ([Please click here to Add Answer](#))

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Two cards are drawn at random from an ordinary deck of 52 cards. Find the probability that both are spade.

Answer ([Please click here to Add Answer](#))

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Made by: Waqar Siddhu

Determine whether the following graph has Hamiltonian circuit, justify your answer.

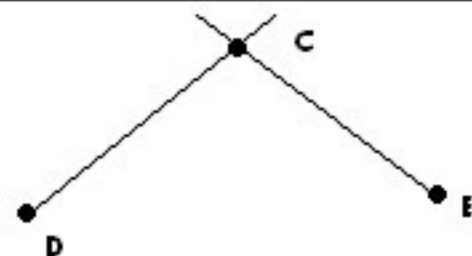


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Draw a binary tree to represent the following expression

$$a/(b-c.d)$$

Answer ([Please click here to Add Answer](#))

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Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{2x+1}{2x+2}$. Is f onto?

Answer ([Please click here to Add Answer](#))

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Find the GCD of 1075,45 using Division Algorithm.

Answer ([Please click here to Add Answer](#))

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Is it possible to have a simple graph with four vertices of degree 1, 1, 3, and 3. If no then give reason? (Justify your answer)

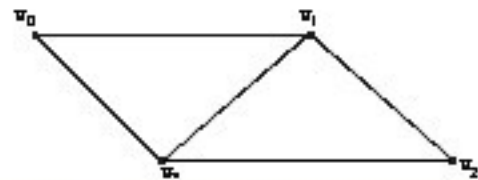
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Find a spanning tree of the graph given below.



Answer ([Please click here to Add Answer](#))

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Let f is defined recursively by $f(0) = 3, f(n+1) = 2f(n) + 2$ then find $f(2)$.

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Answer ([Please click here to Add Answer](#))



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List all 2-combinations for the set {a,b,c}.

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Answer ([Please click here to Add Answer](#))



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Find $P(A|B)$ where

$$P(A) = \frac{1}{2}, P(B) = \frac{1}{3} \text{ and } P(A \cap B) = \frac{1}{4}.$$

Answer ([Please click here to Add Answer](#))

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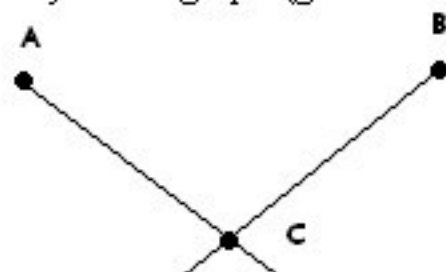
Out of five people in an office, just 3 are to be selected to go to an exhibition. In how many ways can the three be chosen?

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Check the regularity of the graph (given below) and Justify your answer.

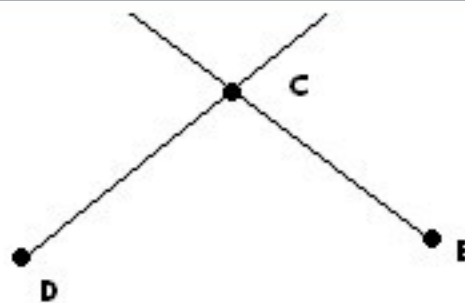


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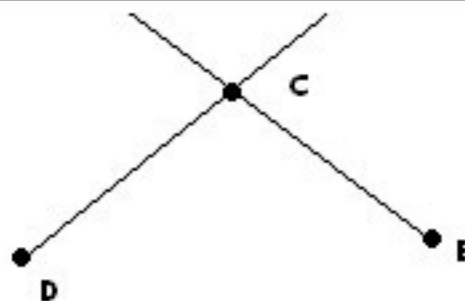


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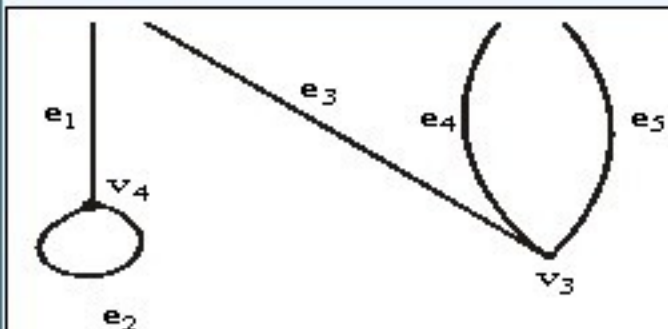


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Answer (Please [click here](#) to Add Answer)



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Suppose that R is a symmetric relation on a set A . Prove that complementary relation $\bar{R}$ is also symmetric.

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Answer ([Please click here to Add Answer](#))



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Find the GCD of 255561, 25 using Division Algorithm.

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Answer ([Please click here to Add Answer](#))



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A pair of fair dice is thrown. Find the probability P that the sum is 10 or greater if

- (i) 5 appear on first die.
- (ii) 5 appear on at least one die.

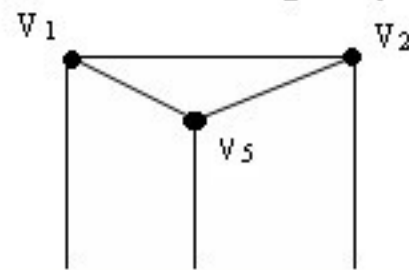
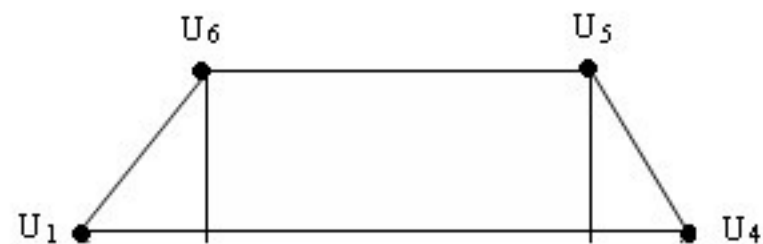
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Answer ([Please click here to Add Answer](#))



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Determine whether the graphs given below are isomorphic. Justify your answer. Also discuss the regularity of both graphs.

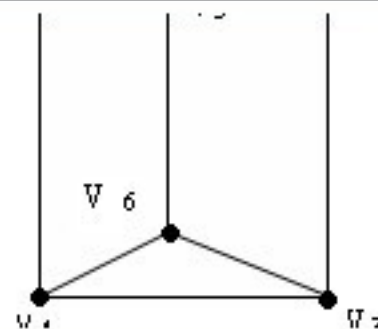
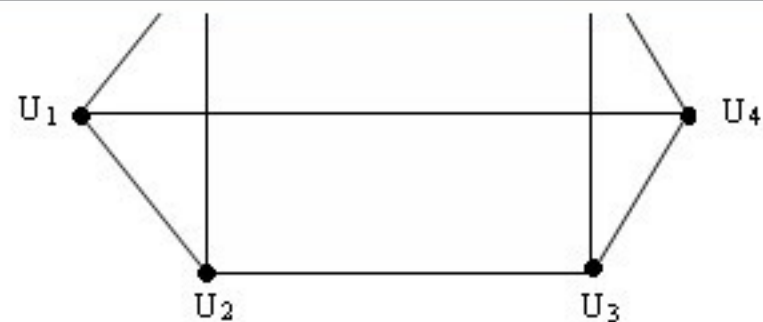


Answer ([Please click here to Add Answer](#))

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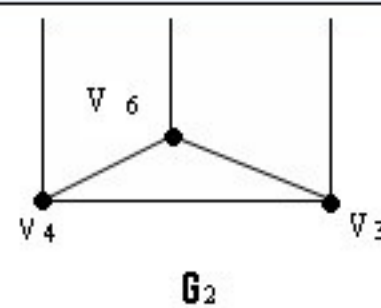


Answer ([Please click here to Add Answer](#))

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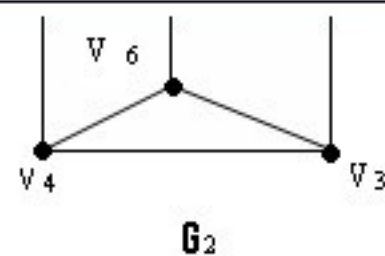
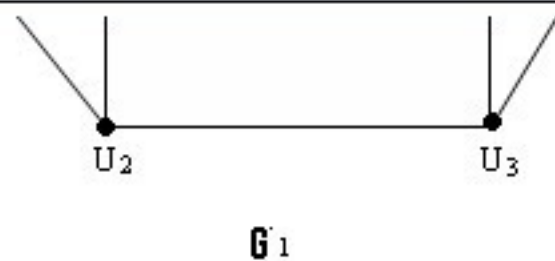


Answer ([Please click here to Add Answer](#))

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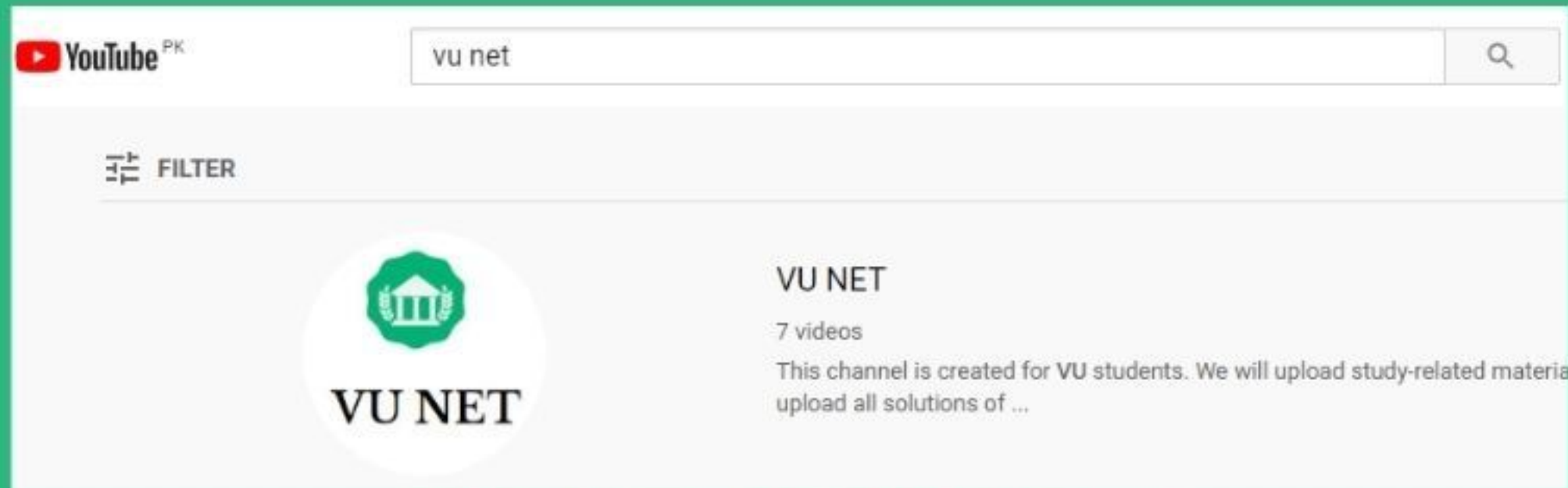
Answer (Please [click here](#) to Add Answer)

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